FACO FLEET

ENTERPRISE FLEET OPERATIONS SYSTEM

Official User Manual & Operating Guide

A comprehensive operating reference covering vehicle onboarding, document compliance, windshield QR code checkpoint verification, operational expense logging, executive approvals, maintenance scheduling, and offline-first cloud synchronization.

DOCUMENT CONTROL & PUBLICATION METADATA

System Version: Faco Fleet Enterprise v0.2.0

Platforms: Desktop (macOS / Windows), Web, Mobile (Capacitor)

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Target Audience: Executives (CEO), Fleet Managers, Secretaries

Classification: Operational Business Document

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System Overview

Faco Fleet is an enterprise-grade fleet management system designed for transport companies operating in environments with variable power and network connectivity. The software incorporates a local-first SQLite engine paired with automatic Turso cloud synchronization, ensuring uninterrupted data entry in remote depots while centralizing executive oversight.

Pillar	Core Functionality	Business Impact
Local Durability	Embedded SQLite database handles all writes locally when internet drops.	Zero downtime during field or depot power outages.
Cloud Coordination	Automatic Turso cloud sync and Cloudinary document storage.	Executives monitor multi-branch operations in real-time.
Field Verification	Public QR stickers readable by any standard smartphone camera.	Eliminates delays at road safety and security checkpoints.

1. System Architecture & Access Control

Access to Faco Fleet is strictly governed by Role-Based Access Control (RBAC). Each employee account is assigned a defined system role ensuring separation of duties between depot staff submitting expenses and executive leadership authorizing disbursements. Drivers are managed operational entities assigned to vehicles and do not hold direct login credentials.

Role	System Title	Primary Responsibilities
CEO	Superadmin	Full executive authority. Configures system users, sets role permissions, manages branches, and authorizes expenses.
FLEET MANAGER	Admin	Operational manager. Registers vehicles and drivers, assigns assets, approves maintenance and expenses, and exports reports.
SECRETARY	Branch Staff	Depot data clerk. Records operational expenses, logs maintenance service requests, and uploads compliance documents.

System Permission Matrix by Role

Action / Operational Domain	CEO (Superadmin)	Manager (Admin)	Secretary
View Fleet Vehicles & Drivers	Yes	Yes	Yes
Register Vehicle / Driver	Yes	Yes	No
Edit Vehicle / Driver Details	Yes	Yes	No
Record Operating Expense	Yes	Yes	Yes
Approve / Reject Expense	Yes	Yes	No
Record Maintenance Request	Yes	Yes	Yes
Approve Maintenance Request	Yes	Yes	No
Upload & View Documents	Yes	Yes	Yes
Export Reports & Audit Logs	Yes	Yes	No
Manage System Users & Roles	Yes	No	No

2. Fleet Registry & Vehicle Onboarding

Every commercial vehicle in the fleet must be onboarded through the Registry. Registration establishes the unique vehicle code, links the registration plate to the VIN, and initializes compliance tracking for insurance and roadworthiness certifications.

Step	Field / Requirement	Operating Guidance
1. Navigation	Sidebar -> Vehicles -> Register Vehicle	Open Fleet Registry and click the '+ Register New Vehicle' button.
2. Identity & VIN	Code, Name, Plate & Chassis VIN	Assign unique fleet code (e.g. TRK-01), name, registration plate, and chassis VIN.
3. Specifications	Make, Model, Year, Colour, Fuel	Record manufacturer specs (e.g. Toyota Hilux 2023, Diesel) and initial operating status.
4. Visual Inspection	Vehicle Photos (Up to 4 Images)	Upload inspection photos documenting exterior condition before fleet deployment.
5. Assignment & Docs	Driver Linking & Certificate Upload	Open vehicle detail view to link a driver; upload insurance/roadworthiness in Documents.

DOCUMENT EXPIRY & COMPLIANCE TRACKING

Vehicle insurance policies, roadworthiness certificates, and carriage permits are tracked in the Documents module (Sidebar -> Documents). Document validity is evaluated automatically, surfacing clear status indicators: active certificates display standard expiration dates, documents expiring within 30 days show an amber warning badge, and expired certificates are immediately flagged with a high-visibility red badge.

3. Windshield QR Verification System

The Windshield QR Verification System provides instant, tamper-proof verification of commercial vehicles at highway checkpoints, company gates, and customer delivery yards without requiring checkpoint officers to hold user accounts or login credentials.

Component	Technical Mechanism	Operational Benefit
Printable Sticker	Standard laminated QR label generated directly from the vehicle detail view.	Affixed to the bottom-left of the windshield for high visibility.
Checkpoint Scan	Standard camera app on any iOS or Android phone resolves the QR code.	Zero app installation required for police or road safety officers.
Public Verification	Opens https://facotransport.com/qr/[code] over secure HTTPS.	Displays live vehicle status, assigned driver details, and driver license expiry.

Step-by-Step QR Printing & Deployment

Step 1 Open Fleet Registry -> Select vehicle -> Click 'QR Card'.

Step 2 Click 'Print QR Badge' to produce an official standardized A6/A5 label.

Step 3 Laminate the printed badge against moisture, UV sunlight, and dust.

Step 4 Affix to the vehicle windshield interior next to inspection decals.

4. Operational Expenses & Multi-Level Approvals

To prevent ghost receipts and unauthorized expenditures, Faco Fleet enforces a rigid submission-and-approval workflow. Operational costs logged by branch secretaries remain in a PENDING state until officially authorized by a Fleet Manager or CEO.

Category	Operational Description	Common Items
REPAIRS	Mechanical, electrical, and breakdown services	Engine overhaul, radiator leak, suspension
PARTS	Physical vehicle replacements and consumables	Tires, brake pads, battery, wiper blades
INSURANCE	Vehicle insurance policy premium payments	Annual comprehensive and third-party coverage
REGISTRATION	State vehicular registration fees	Vehicle renewal stickers, state registry fees
LICENSING	Interstate permits, road taxes and toll passes	Heavy vehicle permits, haulage licensing
MISCELLANEOUS	Other depot and operational trip expenditures	Depot supplies, parking fees, washing

Executive Approval Workflow

1. Submission Staff member selects vehicle, category, enters amount, description, and date in Expenses -> Add Record.

2. Queueing New expenditure enters PENDING state and surfaces immediately in the Executive Approvals queue.

3. Review Manager or CEO opens Approvals -> Inspects expenditure amount, description, and submitting user.

4. Authorization Executive clicks 'Approve' or 'Reject', immediately updating status and executive analytics.

5. Preventive Maintenance & Service Logs

Maintenance tracking minimizes highway breakdowns, audits garage charges, and extends fleet longevity. Staff log service requests itemizing labour, parts, and servicing vendors, which remain pending until authorized by a Fleet Manager or CEO.

Data Field	Scope & Entry Requirement	Operational Purpose
Vehicle Selector	Choose specific vehicle from active registry	Links maintenance history to unique vehicle code.
Service Description	Detailed repair scope (e.g. Brake Pads & Oil)	Identifies mechanical or electrical work performed.
Labour Cost	Numeric value for mechanical technician labour	Separates service charges from parts costs.
Parts Cost (¦	Numeric value for installed replacement parts	Audits parts pricing and component replacements.
Total Cost (¦	Automated sum of Labour + Parts costs	Directly updates cumulative vehicle expense metrics.
Garage / Vendor	External workshop or mechanic name	Tracks contractor accountability and warranty claims.
Next Service Date	Calendar target date for subsequent checkup	Establishes upcoming service horizon for vehicle.

Maintenance Status & Lifecycle

Status State	Operational Condition	System Impact
PENDING	Logged by Secretary or Manager	Surfaces in Approvals queue; expenditure held pending review.
APPROVED	Authorized by Manager or CEO	Cost approved, updates monthly maintenance KPI summaries.
REJECTED	Denied by leadership	Work order flagged as rejected with audit trail record preserved.
MAINTENANCE	Vehicle registry status	Flagged as under maintenance on dashboard and fleet list.

6. Executive Reports, Excel Exports & Audit Trails

The Reports module generates board-level operational audits, compliance reviews, and financial summaries. All reports are generated on-demand and exported in formatted PDF, Microsoft Excel (.xlsx), or raw CSV for executive presentations and external financial audits.

Report Title	Data Coverage & Fields	Formats
Vehicle Utilization & Master Report	Full fleet inventory, registration numbers, VINs, fuel types, branch locations, and operational statuses.	PDF, Excel, CSV
Driver Assignment Report	Active driver directory, commercial license numbers, license expiry dates, emergency contacts, and assigned vehicles.	PDF, Excel, CSV
Maintenance & Servicing Expense Report	Itemized service logs, labour vs parts breakdown, total costs, repair descriptions, and servicing garage vendors.	PDF, Excel, CSV
Executive Operating Cost Summary	Lifetime vehicle operational expenditures categorized by type (repairs, parts, insurance, licensing) and branch.	PDF, Excel, CSV

System Audit Log Trail

Activity Stream	Surfaced on Executive Dashboard: displays actor, timestamp, action type, and entity details.
Tamper-Proof Logs	Records user logins, vehicle creations, expense submissions, and manager approvals.
One-Click Export	Downloadable as CSV via 'Download CSV' directly on the Executive Dashboard.

7. Offline-First Durability & Background Sync

Faco Fleet employs a local-first SQLite architecture. Unlike web-only systems that freeze when the internet cuts out, Faco Fleet continues operating locally with zero disruption.

Indicator State	System Condition	Operating Rule
Cloud Synced (Green)	Active Internet Connection	Local records match Turso cloud database in real-time.
Pending Push (Yellow)	Connection Restored	Local changes queued and actively awaiting background sync.
Offline (Amber)	Internet Disconnected	Records save locally to SQLite. Staff continue working with zero downtime.
Syncing... (Blue)	Reconnection in Progress	Background queue pushes local entries and pulls cloud updates.
Conflicts (Red)	Dual-Device Edit Conflict	Flagged with direct resolution link to Settings -> Sync Conflicts.

Local-First Architecture Mechanics

Local SQLite Database	Embedded database engine in desktop and mobile app. Handles all reads/writes locally.
Turso Cloud Sync	Distributed LibSQL cloud database. Synchronizes depot branches with executive headquarters.
Manual Sync Trigger	Users can click the circular refresh icon next to the sync badge in the top header anytime.

8. In-App Walkthrough & Help Center Guide

Faco Fleet features a built-in Help & Operations Center designed to train depot staff and managers directly within the application. Users can launch interactive spotlights, track onboarding milestones, and download this manual without leaving their workspace.

Help Center Feature	Access Point	Operational Capability
Help Drawer	Sidebar -> Help or Top Header '?'	Opens slide-over panel with tours, onboarding checklist, and user manual download.
Interactive Tours	Help Drawer -> Tours Tab	Step-by-step spotlight tours guiding users into live forms (vehicles, expenses, approvals).
Onboarding Checklist	Help Drawer -> Checklist Tab	Tracks essential setup tasks (adding vehicles, assigning drivers, testing offline sync).
Manual Download	Help Drawer -> Download (PDF)	Instant one-click access to this official monochrome operating manual.

Frequently Asked Questions

Q: What if our depot loses power and internet for days?	A: All entries remain safely persisted in the local SQLite database. They will automatically sync to Turso once power and connection are restored.
Q: Can police verify a vehicle at a highway checkpoint?	A: Scanning the windshield QR code with any smartphone camera opens the live HTTPS verification badge displaying vehicle details and driver credentials.
Q: How do we reset user passwords or manage permissions?	A: The CEO can manage user accounts, assign roles, and toggle permissions in Settings -> User Management.
Q: Can we re-run the interactive walkthrough anytime?	A: Yes. Click the Help button in the sidebar or '?' in the top header and select any tour track to replay it.